

Efficient and Scalable Methods for the Acquisition, Retrieval, and Understanding of Open Tabular Data

Ph.D. Defense

Antoine Gauquier

Supervisors: Ioana Manolescu & Pierre Senellart

DI ENS, ENS, CNRS, PSL University & Inria

September 23, 2026



- 1 Motivation: Open Statistical Data
- 2 Overview of Our Approach
- 3 Efficient Focused Web Crawling for Scalable Data Acquisition
- 4 Efficient and Scalable Search for Statistics
- 5 Conclusions and Perspectives

- 1 Motivation: Open Statistical Data
- 2 Overview of Our Approach
- 3 Efficient Focused Web Crawling for Scalable Data Acquisition
- 4 Efficient and Scalable Search for Statistics
- 5 Conclusions and Perspectives

Statistics Are at the Heart of Public Debate

Citizens



Has my **salary** kept up with **inflation** since 2019?

Is **unemployment** really **rising** since 2025?

Is **air pollution** worse in Paris than in Lyon?

Journalists



What was the value of **EU exports** to the US in 2023?

What is the proportion of **dissaving households** in the European Union of 27 for 2015?

What is the proportion of **people aged 15 to 24** in France in 2021?

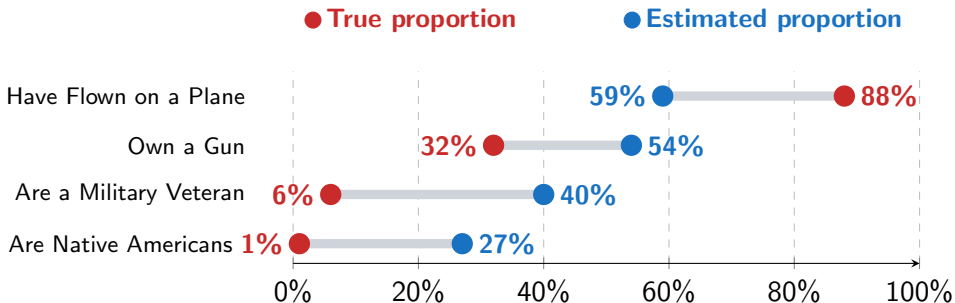
Employment • Education • Health • Demography • Energy • Science • Environment • Transport • Tourism...

Yet Public Perception is Often Wrong

Large study [Ort22] on **misperception** of group sizes by Americans:

- Minority groups are usually largely overestimated
- Most majority groups are also underestimated

Examples:



The Need for Statistical Data

Statistical data are essential for:

- ① disseminating **reliable information** to citizens,
- ② countering the spread of **misinformation**,
- ③ supporting the work of **journalists**.

The ever-growing volume of information calls for **automatic methods** to **find and access this data**.

Who Produces These Statistics?

Different **providers** publish statistics:

- *National agencies*: INSEE (France), Labor/Census Bureaus (US), ABS (AU) → statistics professionals
- *International bodies*: Eurostat, IMF, WB, UN Agencies
- *Smaller entities*: Trade unions, NGOs → publish **highly-specialized data**

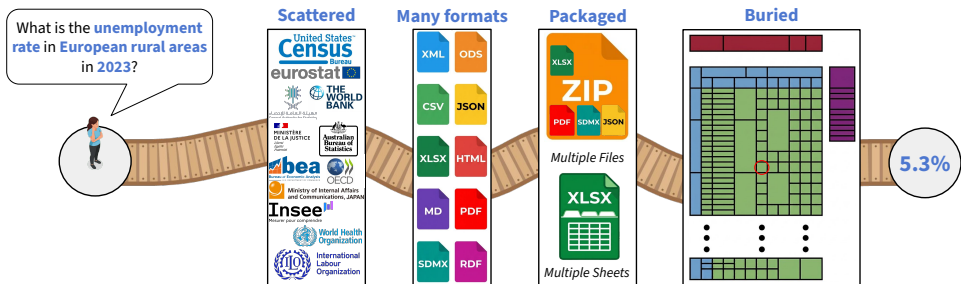


Trustworthy statistics are scattered across hundreds of providers.
Their **accessibility and readability** vary greatly.

Why Is It So Hard? Because Available $\neq$ Usable!

The difficulty is not that the data does not exist, it is finding:

- the right **figure**,
- at the right **place and time**,
- in the right **table**,
- in the right **data file**,
- in the right trustworthy **source**.



Time-consuming and painful, and must be carried **for each single query!**

Why Not Just Google It?

Search engines are built to find **relevant online resources** from a query.

Two problems:

- 1 At best: give the **right file in the right source** → do not resolve **packaged** nor **buried** data

Why Not Just Google It?

Search engines are built to find **relevant online resources** from a query.

Two problems:

- ① At best: give the **right file in the right source** → do not resolve **packaged** nor **buried** data
- ② and even for that, **they fail in practice**:

Service	Query	Returned	Actually exist
Google Search capped at 1 000 results	Justice · PDF	302	> 9 000
	Interior · XLS	38	1 546
	Justice · TSV	not recognized	11 097
	ILO · ZIP	0	> 2 239
Google Dataset Search	Justice	109	1 188
	ILO	93	> 170 000
	Census	312	> 800 000

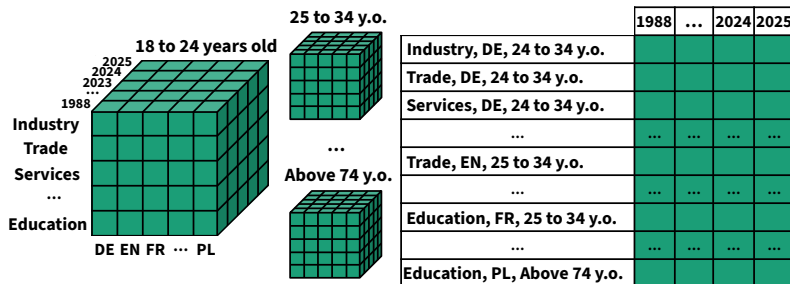
Why Classical Table Discovery Is Insufficient

Table Discovery (TD) systems expect **relational** tables: a clean schema, one row per entity

Statistical data are **multidimensional aggregates**, **serialized** into 2D:

- No schema, cells **overwhelmingly numerical** (meaning only in **headers**)
- Up to **millions of rows**, across **thousands of tables** → classical TD systems do not scale (*detailed later*)

Number of new jobs created across the European Union

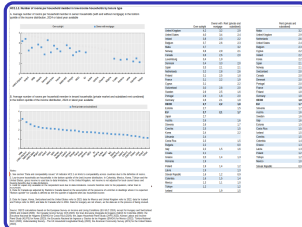


Raw Files Are Not Clean Tables

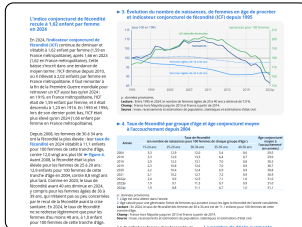
easy

```
<table id="produit-tableau-figures" class="tableau-produit" style="caption:Nombre moyen d'enfants à divers âges par génération" >thead>
<tr class="">
<th class="" rowspan="2" scope="col">
<span class="infobulle-accessible adhoc-definition" Génération
</span>
</th>
<th class="" colspan="9" scope="col">Âge atteint à la
</tr>
<tr class="">
<th class="" scope="col">20 ans </th>
<th class="" scope="col">22 ans </th>
<th class="" scope="col">24 ans </th>
<th class="" scope="col">26 ans </th>
<th class="" scope="col">28 ans </th>
<th class="" scope="col">30 ans </th>
<th class="" scope="col">35 ans </th>
<th class="" scope="col">40 ans </th>
<th class="" scope="col">50 ans </th>
</tr>
</thead>
```

Extraction difficulty



hard



HTML / DB exports

Explicit structure

Trivial extraction

Spreadsheets

Most frequent in practice

Structured yet free-form

Made for human eyes

PDF

No explicit structure

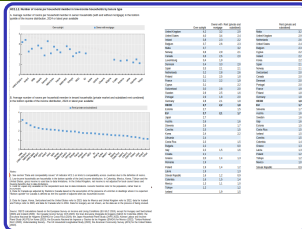
Hardest to extract

Raw Files Are Not Clean Tables

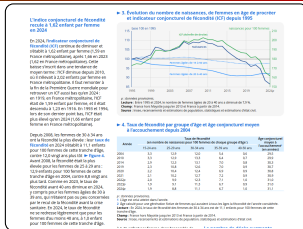
easy

```
<table id="produit-tableau-figures" class="tableau-produit" sty
<caption>Nombre moyen d'enfants à divers âges par génération
<thead>
<tr class="">
<th class="" rowspan="2" scope="col">
<span class="infobulle-accessible adhoc-definition">
Génération
</span>
</th>
<th class="" colspan="9" scope="col">Âge atteint à la
</tr>
<tr class="">
<th class="" scope="col">20 ans </th>
<th class="" scope="col">22 ans </th>
<th class="" scope="col">24 ans </th>
<th class="" scope="col">26 ans </th>
<th class="" scope="col">28 ans </th>
<th class="" scope="col">30 ans </th>
<th class="" scope="col">35 ans </th>
<th class="" scope="col">40 ans </th>
<th class="" scope="col">50 ans </th>
</tr>
</tr>
```

Extraction difficulty



hard



HTML / DB exports

Explicit structure

Trivial extraction

Spreadsheets

Most frequent in practice

Structured yet free-form

Made for human eyes

PDF

No explicit structure

Hardest to extract

Challenges:

Many languages

Free & varied layouts

1+ Tables per sheet

Merged cells, sparse regions

Variable levels of metadata

Thousands of tables, millions of cells

Outline

- 1 Motivation: Open Statistical Data
- 2 Overview of Our Approach
- 3 Efficient Focused Web Crawling for Scalable Data Acquisition
- 4 Efficient and Scalable Search for Statistics
- 5 Conclusions and Perspectives

Overview of Our Approach

Statistical Data is Hard to Leverage

From the motivation, statistical data is **available yet hard to exploit**:

- Scattered, packaged, and buried in heterogeneous sources
- Multidimensional aggregates that classical TD cannot handle at scale
- Hard to get clean tables from raw data → especially spreadsheets

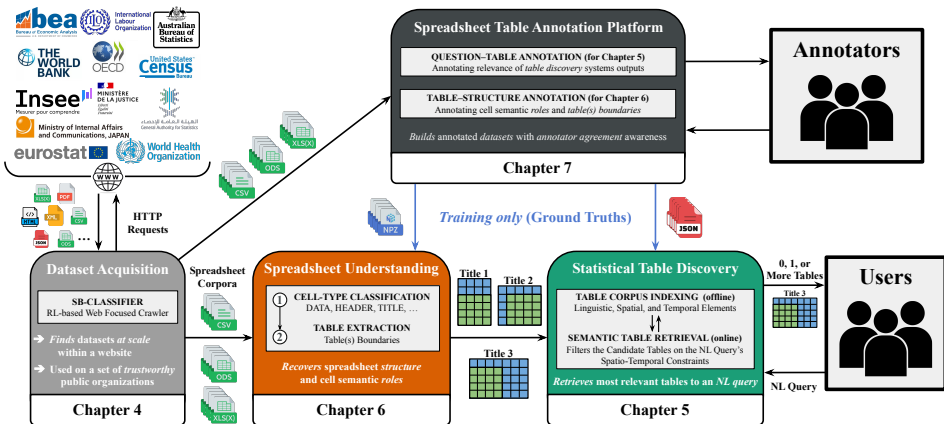
Research Question

How can we **acquire**, **understand**, and **retrieve** open tabular data **efficiently** and **at scale**?

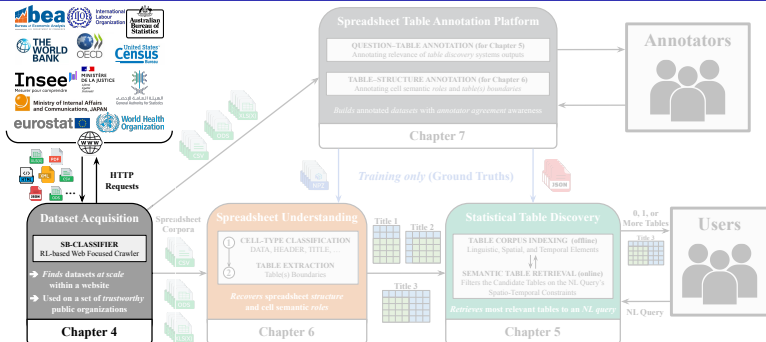
Our Answer: An End-to-End Statistical Data Pipeline

- 1 Efficiently **acquire** all data files from a trusted Web source
- 2 **Recover** the table structure and cell roles in spreadsheets
- 3 Efficiently **query** corpora of statistical tables in natural language

An End-to-End Statistical Data Pipeline



Dataset Acquisition Problem

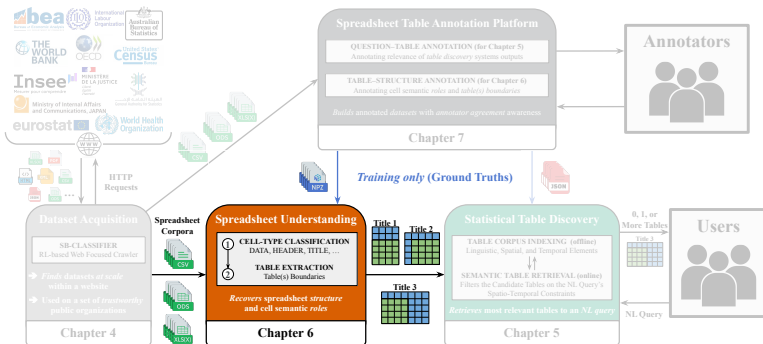


Problem: Collect as many data files as possible within a trusted website, while minimizing requests and exchanged data

- An **RL-based crawler** learning the fruitful parts of the website
- On large websites: **90% of targets** while visiting only **20% of pages**

EDBT 2026 [GMS26a], extension under review [GMS26b], BDA 2024 [GMS24b], and Infox sur Seine 2024 [GMS24a]

Spreadsheet Understanding Problem

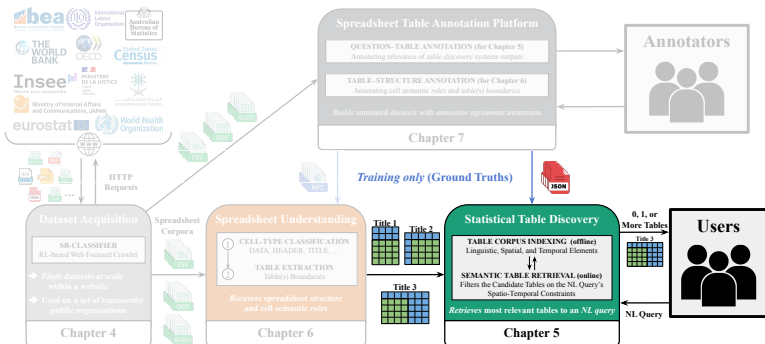


Problem: Recover machine-readable tables (cell roles and table boundaries) from spreadsheets

- Classifier + CRF for cell types, followed by deterministic extraction
- Runs on a single machine, no GPU, competitive with heavier systems

CIKM 2026 [GMS26c] and BDA 2026 [GMS26d]

Statistical Table Discovery (STD) Problem

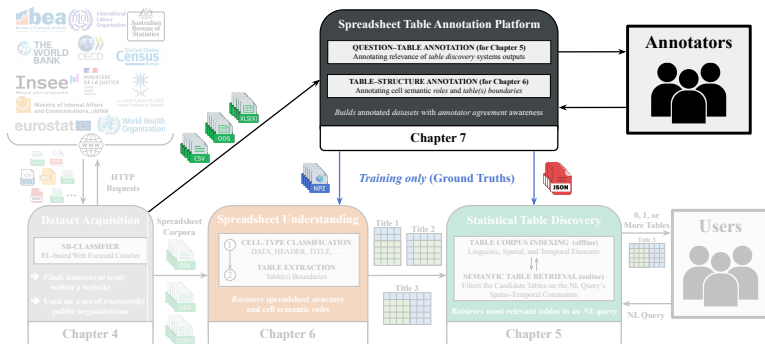


Problem: Given a natural-language question, retrieve the relevant tables among a corpora of thousands of large, multidimensional statistical tables.

- **STAR:** Scales to large corpora, **no LLM**, **no GPU**
- **Outperforms much heavier** general table discovery systems

ICDE 2026 [GEG⁺26a], TLDKS [GEG⁺26b], and BDA 2025 [GEG⁺25]

Annotation Support



Problem: Annotating real spreadsheets is hard and annotators can disagree

- **TABAGREE**: a multi-annotator platform for spreadsheet annotation
- Treats **inter-annotator agreement** as an interpretable signal
- **Produced the datasets** used for STD and spreadsheet understanding

Demonstration paper [GMS26e] to be submitted, and BDA 2026 [GMS26f]

Guiding Principles Throughout This Thesis

- ① **Efficiency** under real constraints (carried by the targeted users)
→ Modest hardware, *without GPU*
- ② **Heterogeneity**
→ Built and tested to handle a variety of *formats, topics, languages, website structures*...
- ③ **Scalability**
→ Designed to ingest *large volumes* (up to *tens of thousands of tables*)
- ④ **Reproducibility**
→ *Open-source code* and two *publicly released datasets* (STATSHEETS, and STAR's one)
- ⑤ **Interpretability**, as much as possible

- 1 Motivation: Open Statistical Data
- 2 Overview of Our Approach
- 3 Efficient Focused Web Crawling for Scalable Data Acquisition
- 4 Efficient and Scalable Search for Statistics
- 5 Conclusions and Perspectives

Acquiring Data at Scale Is Hard

Websites can be **huge** with only a **small fraction** of published data files

Exhaustively crawling those websites takes a lot of **space** and most importantly **time**:

- Taken by the crawl of a lot of “uninteresting” resources
- **Crawling ethics** requires to wait between two HTTP requests
→ For a website of **1M pages**: roughly **11 days** of waiting

Generic crawler for any given website (selected beforehand) → we **cannot make any assumption about the website's structure/content**.

Problem Statement

Problem: Given the starting URL of a website, we want to **retrieve** as many **targets** as possible, while **minimizing** the **resource** consumption.

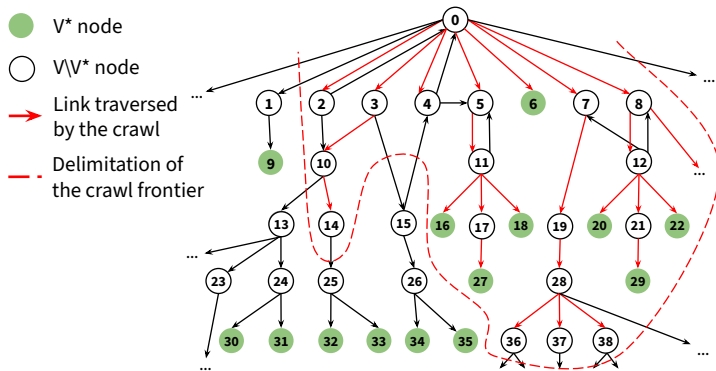
- A **target**: **file uploaded** on the website, satisfying some constraints specified by the user
→ For Statistical Dataset (SD) acquisition: CSV, Excel, PDF...
- **Resources**: number of **HTTP requests** sent to the server, and **data volume** exchanged with the server

Can this problem be solved **exactly**?

Formalization: The Graph Crawling Problem

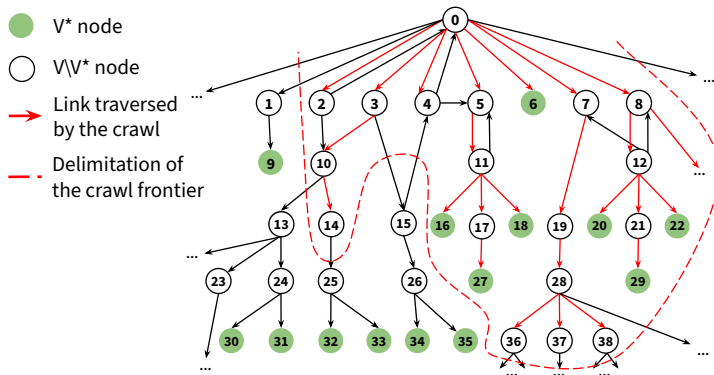
Website graph: $G = (V, E, r, \omega, \lambda)$, with:

- V a finite set of *vertices* (webpages)
- $E \subseteq V^2$ a set of *edges* (hyperlinks)
- $r \in V$ the *root* of the graph (input webpage; 0 here)
- $\omega : V \rightarrow \mathbb{R}^+$ a *cost* function (cost of retrieving that page)
- $(\lambda : E \rightarrow \mathcal{L}$ a *labeling* function).



Formalization: The Graph Crawling Problem

Given a website graph $G = (V, E, r, \omega, \lambda)$ and a subset V^* of targets of V , the graph crawling problem is to **find a crawl** $T = (V', E')$ of G **with** $V^* \subseteq V'$, **of minimal total cost**.



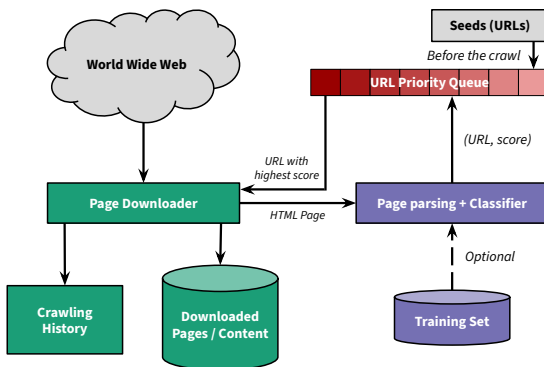
NP-hard problem $\rightarrow$ calls for a **heuristic** yet **effective** solution

Existing Solutions: Focused Crawlers

Search engines are **poorly suited**, and exhaustive crawlers are **too costly**.

Focused Crawlers:

- Prioritize pages following a **focus** (an objective).
- A **classifier** (usually) assesses the **likelihood** that an hyperlink leads to a page part of the focus.



Existing Solutions: Focused Crawlers

Two kind of focused crawlers:

- **Topical**: focus is a predefined **topic** $\rightarrow$ keywords, documents. . .
- **Non-Topical**: rarer; different focus:
 - ANTHELION [MMB14]: semantic annotations in Web pages
 - ACEBOT [FS15]: as much diverse textual content as possible

Existing Solutions: Focused Crawlers

Two kind of focused crawlers:

- **Topical**: focus is a predefined **topic** $\rightarrow$ keywords, documents. . .
- **Non-Topical**: rarer; different focus:
 - ANTHELION [MMB14]: semantic annotations in Web pages
 - ACEBOT [FS15]: as much diverse textual content as possible

Topical crawlers are **ill-suited** for our problem

- We want targets about **any possible subject** covered by the website
- Defining a topic $\implies$ choosing **one (or several) languages**

Best approach: non-topical focused crawlers

Research Hypotheses

- 1 Hyperlinks similarly structured in the HTML page were they were found lead to similar content.
- 2 It is possible to learn which hyperlinks are most likely to lead to targets, given their link structure.

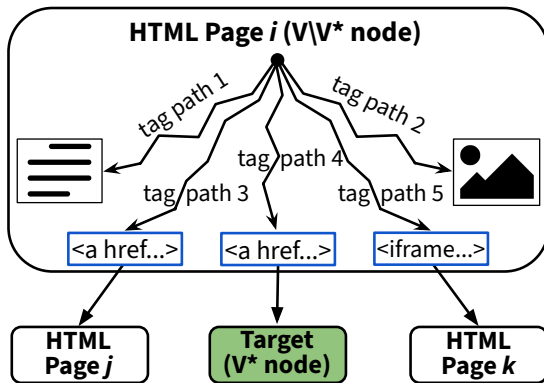
Idea: We can separate, for a given website, parts of it that are rich in targets, from ones that are not:

- without prior knowledge of the website's structure/content
- just assuming existence of some structure

Building Groups of Hyperlinks

How to do it: represent hyperlinks with their associated **tag path**

They are computed from the HTML page's **Document Object Model**:



Building Groups of Hyperlinks

We form **groups of hyperlinks** with a merging strategy based on **similarity** between the tag paths...

...to **separate the fruitful groups** from the less interesting ones, as we assume that **similar hyperlinks** will lead to similar kind of content.

```
https://opm.justice.gouv.fr/content/notice-dépôt-dune-candidature-au-concours-de-greffier-de-tribunal-de-commerce-annexe-1-0
```

```
/html/body.html front not-logged-in one-sidebar sidebar-  
first page-node/div.jumbottron opm-content-container/  
div.container/div.row/div.col-sm-3 well-resp/div.block  
block-block last odd#block-block-7/p/a
```

```
https://www.justice.gouv.fr/lettre-direction-affaires-civiles-du-sceau
```

```
/html/body.html front not-logged-in one-sidebar sidebar-  
first page-node/div.jumbottron opm-content-container/  
div.container/div.row/div.col-sm-3 well-resp/div.block  
block-block first odd#block-block-6/p/a
```

```
https://opm.justice.gouv.fr/content/notice-dépôt-dune-candidature-au-concours-de-greffier-de-tribunal-de-commerce
```

```
/html/body.html front not-logged-in one-sidebar sidebar-  
first page-node/div.jumbottron opm-content-container/  
div.container/div.row/div.col-sm-3 well-resp/div.block  
block-block last odd#block-block-7/p/a
```

```
https://www.justice.gouv.fr/grands-dossiers/justice-amiable
```

```
/html/body.path-frontpage page-node-type-homepage/  
[...] /div.fr-col-md-6 fr-col-12/article.fr-enlarge-  
link fr-card fr-card-detail-sm fr-card--horizontal/  
div.fr-card_body/div.fr-card_content/h3.fr-card_t  
itle fr-title/a.fr-card_link
```

```
https://www.justice.gouv.fr/prise-charge-victimes-dinfractions-penales
```

```
/html/body.path-frontpage page-node-type-homepage/  
[...] /article.fr-enlarge-link fr-card fr-card-  
detail-sm fr-card--horizontal/div.fr-card_body/  
div.fr-card_content/h3.fr-card_titlefr-title/  
a.fr-card_link
```

```
https://www.justice.gouv.fr/plan-daction-justice
```

```
/html/body.path-frontpage page-node-type-homepage/  
[...] /div.fr-col-md-6 fr-col-12/article.fr-enlarge-  
link fr-card fr-card-detail-sm fr-card--horizontal/  
div.fr-card_body/div.fr-card_content/h3.fr-card_t  
itle fr-title/a.fr-card_link
```

Learning Fruitful Groups: Reinforcement Learning

How can we dynamically separate fruitful groups from other ones?

Learning Fruitful Groups: Reinforcement Learning

How can we dynamically separate fruitful groups from other ones?

→ Reinforcement Learning (RL).

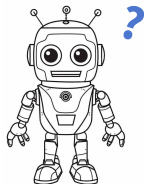
Leveraging Sleeping Multi-Armed Bandits [KNS10]

Learning Fruitful Groups: Reinforcement Learning

How can we dynamically separate fruitful groups from other ones?

→ Reinforcement Learning (RL).

Leveraging Sleeping Multi-Armed Bandits [KNS10]

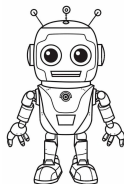


Learning Fruitful Groups: Reinforcement Learning

How can we dynamically separate fruitful groups from other ones?

→ Reinforcement Learning (RL).

Leveraging Sleeping Multi-Armed Bandits [KNS10]



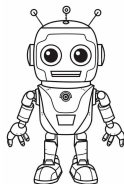
Selects a **group**

Learning Fruitful Groups: Reinforcement Learning

How can we dynamically separate fruitful groups from other ones?

→ Reinforcement Learning (RL).

Leveraging Sleeping Multi-Armed Bandits [KNS10]



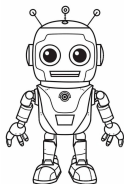
Selects a **group**
Follows a hyperlink from it

Learning Fruitful Groups: Reinforcement Learning

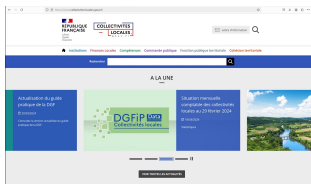
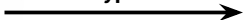
How can we dynamically separate fruitful groups from other ones?

→ Reinforcement Learning (RL).

Leveraging Sleeping Multi-Armed Bandits [KNS10]



Selects a **group**
Follows a hyperlink from it

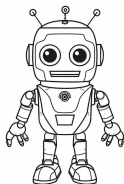


Learning Fruitful Groups: Reinforcement Learning

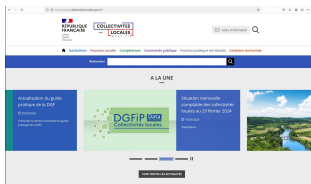
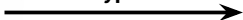
How can we dynamically separate fruitful groups from other ones?

→ Reinforcement Learning (RL).

Leveraging Sleeping Multi-Armed Bandits [KNS10]



Selects a **group**
Follows a hyperlink from it



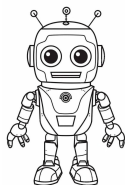
How many **targets**
can we reach
from this page?

Learning Fruitful Groups: Reinforcement Learning

How can we dynamically separate fruitful groups from other ones?

→ Reinforcement Learning (RL).

Leveraging Sleeping Multi-Armed Bandits [KNS10]



Selects a **group**
Follows a hyperlink from it

Gets a **reward**:
Score describing **how**
“good” was this choice



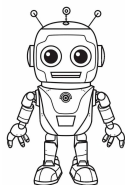
How many **targets**
can we reach
from this page?

Learning Fruitful Groups: Reinforcement Learning

How can we dynamically separate fruitful groups from other ones?

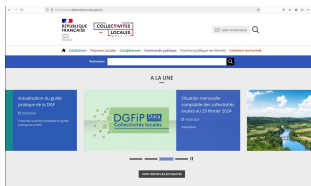
→ Reinforcement Learning (RL).

Leveraging Sleeping Multi-Armed Bandits [KNS10]



Selects a **group**
Follows a hyperlink from it

Gets a **reward**:
Score describing **how**
“good” was this choice



How many **targets**
can we reach
from this page?

Reward: Number of reachable, unobserved targets from the HTML page.

Exploration–Exploitation Trade-Off

At each crawling step, the choice of the group is a trade-off between:

- **Exploitation** of groups so far identified as fruitful
- **Exploration** of few times selected groups, that might hide fruitful hyperlinks.

Works well!

Exploration–Exploitation Trade-Off

At each crawling step, the choice of the group is a trade-off between:

- **Exploitation** of groups so far identified as fruitful
- **Exploration** of few times selected groups, that might hide fruitful hyperlinks.

Works well!

But requires to tell, **just given an URL**, if it leads to a target or not:

- `https://www.assemblee-nationale.fr/14/pdf/ta/ta0686.pdf`
→ Easy
- `https://www.education.gouv.fr/media/90068/download`
→ Manageable
- `https://www.justice.gouv.fr/en/node/9961`
→ Requires learning

Reward Computation

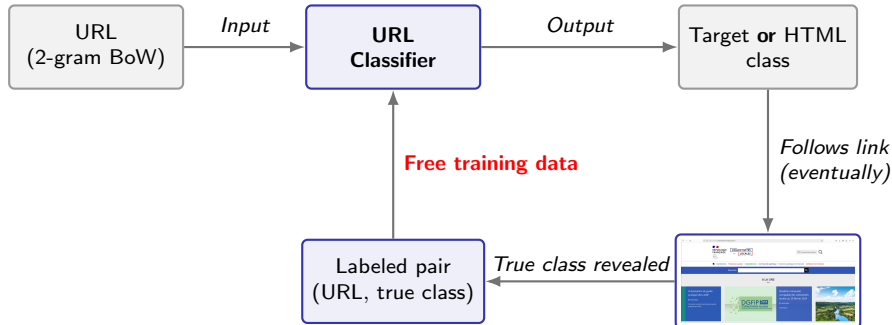
How to compute the reward?

Naive way: HTTP HEAD requests over the URL to get Content-Type

→ **Too costly:** exactly what we want to avoid

Solution: actively train a **URL classifier**

→ Label its own training data **as the crawl proceeds**



Dataset Overview

Scope:

- **18 websites** expected to host **statistical datasets**
- **> 22 millions** distinct pages collected

Diversity:

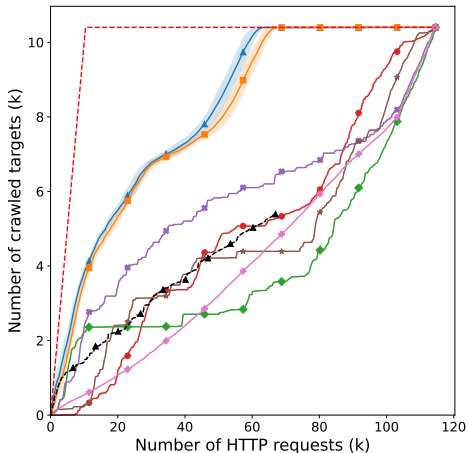
- Monolingual & multilingual websites (>20 languages)
- Websites can be:
 - fully crawled
 - partially crawled (limits: 1 million visited pages **or** 3 weeks)



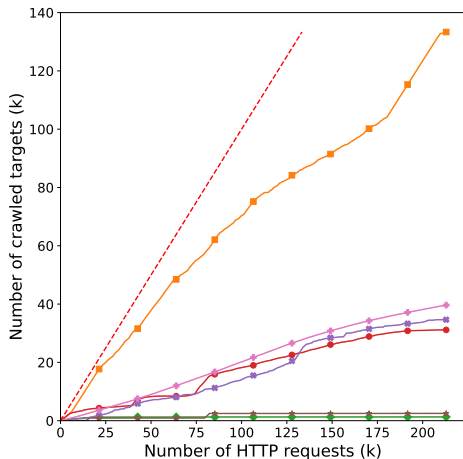
Results Overview



French Ministry of Education



World Health Organization



Outline

- 1 Motivation: Open Statistical Data
- 2 Overview of Our Approach
- 3 Efficient Focused Web Crawling for Scalable Data Acquisition
- 4 Efficient and Scalable Search for Statistics**
- 5 Conclusions and Perspectives

Querying Statistical Tables

Assume the previous steps are solved:

- Data files have been **acquired**
- Table structure has been **recovered**, with metadata extracted (e.g., the title)

→ we now have corpora of **(tens of) thousands** of statistical tables

We need methods to query them that are:

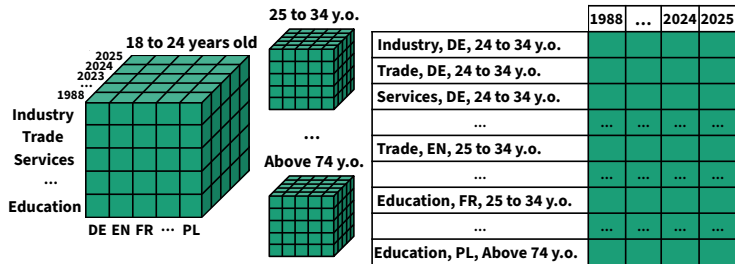
- 1 **Automatic** and **scalable**
- 2 Driven by **natural-language** queries

Statistical vs Relational Tables

Recall: Statistical tables represent **multidimensional data**:

- E.g., “New jobs created” by year, country, sector, age group: each statistical value (*metric*) is defined for a set of d dimension values.
- Flattened into 2D tables for compatibility with classical tools: **serialization**.

Number of new jobs created across the European Union

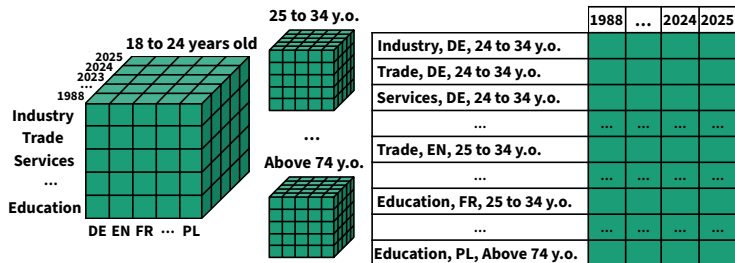


Statistical vs Relational Tables

Challenges of serialized statistical tables:

- Semantics are in **row or column headers**
- High dimensionality leads to **very large tables**
- **Context** for dimension values often missing

Number of new jobs created across the European Union



Can traditional **TD systems** handle this complexity?

Existing Table Discovery Methods

Recall: Table Discovery (TD) $\rightarrow$ finding relevant table(s) for a query

Recent methods [WF23, BAW⁺25, GHM⁺25] rely on DL and LLMs:

- Annotated data generation
- Table schema synthesis
- Result re-ranking

$\rightarrow$ designed for **relational tables** with well-defined schemas

This leaves **two gaps** for statistical data:

- 1 They do not handle **Statistical Table Discovery** (STD): TD over multidimensional, mostly numerical tables
- 2 They are **resource-heavy** (LLMs, GPUs) $\rightarrow$ unfit for a broad audience

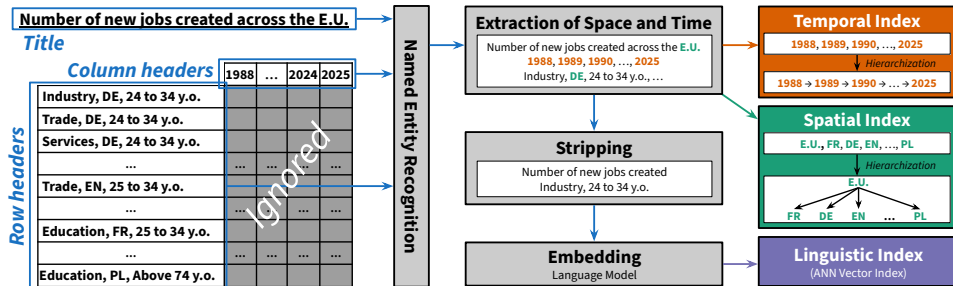
The STAR Approach: Indexing

STAR = Space- and Time-Aware Statistic Retrieval

Indexing along **three axes**:

- **Temporal** and **spatial** values, extracted and indexed **exactly**, in hierarchies
- **Linguistic** content (title + headers), embedded by a language model

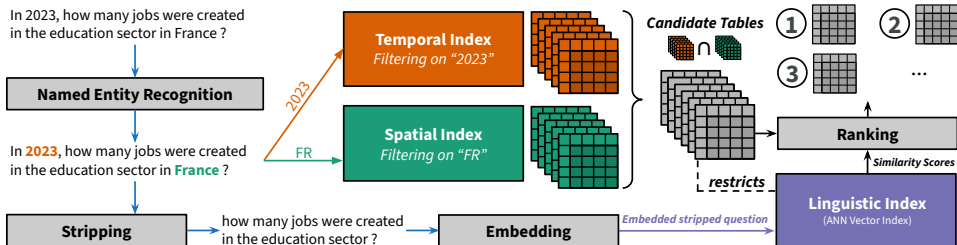
→ **Numerical values are ignored**



The STAR Approach: Querying

A natural-language question is answered in **two steps**:

- **Filter**: keep only tables covering the question's **time** and **location** (candidate tables)
- **Rank**: score the candidates by **linguistic** similarity



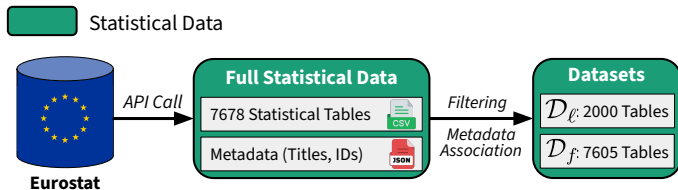
Experiments: Datasets

Dataset: 7 605 Eurostat tables in CSV format ($\mathcal{D}_f$):

- As **serialized tables**
- With $d \geq 1$ **dimensions**: always includes time, and in most cases space
- Along with their **titles** (and IDs)

$\mathcal{D}_\ell$: Subset of 2 000 smallest tables, as **some systems do not scale to $\mathcal{D}_f$**

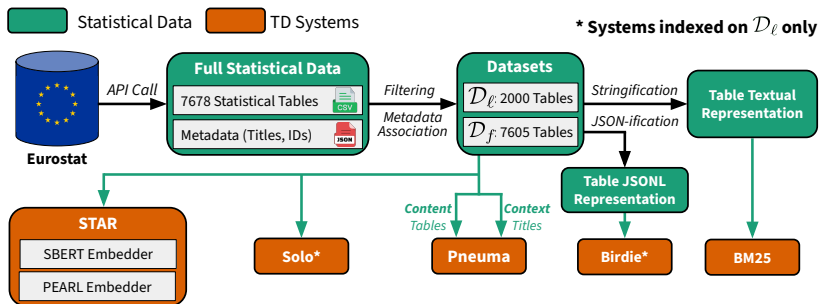
Dataset	#Tables	Size (MB)	#rows	#columns
$\mathcal{D}_f$	7 605	98 521.7	69 077 ($\pm$ 318 706)	41 ($\pm$ 270)
$\mathcal{D}_\ell$	2 000	30.3	111 ($\pm$ 101)	19 ($\pm$ 23)



Experiments: Systems

We evaluate 7 systems:

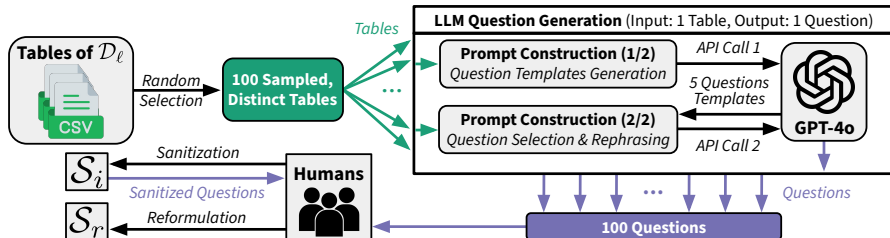
- STAR using standard SBERT embedder [RG19]
- STAR using PEARL [CVS24] → LM for short, context-free phrases
- SoTA TD systems [WF23, BAW⁺25, GHM⁺25]
- BM25 baseline and its variant BM25+Syn.



Experiments: Question Sets

Two **questions sets** (for evaluation):

- **Initial** ($\mathcal{S}_i$): Lexically close to the table content
- **Reformulated** ($\mathcal{S}_r$): Same semantic meaning as $\mathcal{S}_i$, **but** with maximal **lexical variation**



Results in two **high-quality, manually sanitized** question sets
Evaluating on both **lexical** and **semantic** understanding

We score each system with 2 metrics:

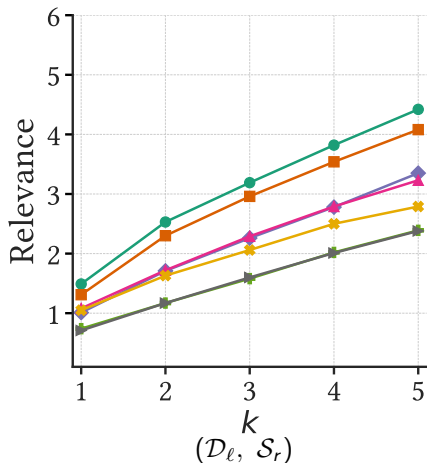
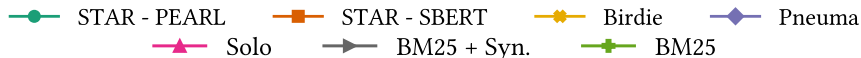
- **HitRate@ k** : Proportion of questions from a question set for which a system returned the *golden* table in its top- k ($1 \leq k \leq 10$).
→ **Standard metric in the literature**
- **Relevance@ k** : measures a system's ability to return semantically relevant tables in its top- k results ($1 \leq k \leq 5$).
→ **More interesting: there might be ≥ 1 relevant tables**

Experiments: Running Time

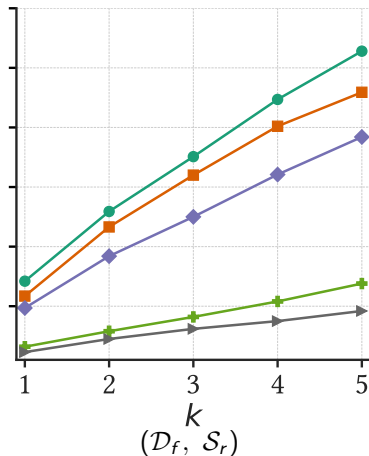
Model	Indexing Time (s)			Query Time (s)		
	$\mathcal{D}_\ell$	$\mathcal{D}_f$	$(\mathcal{D}_\ell, \mathcal{S}_i)$	$(\mathcal{D}_\ell, \mathcal{S}_r)$	$(\mathcal{D}_f, \mathcal{S}_i)$	$(\mathcal{D}_f, \mathcal{S}_r)$
BM25	13	7 701	0.95	0.91	41.83	45.23
BM25+Syn.	13	7 701	3.39	3.34	47.16	51.47
Solo	81 847	OOM	1 267.05	1 143.06	OOM	OOM
Pneuma	45 300	532 020	3 793.23	3 680.90	1 770.44	1 814.24
Birdie	24 135	OOM	20.83	20.32	OOM	OOM
STAR - SBERT	371	45 103	36.77	37.66	186.62	190.95
STAR - PEARL	208	44 279	29.68	34.83	106.19	116.68

- Lexical-only methods scale and are the fastest
- Solo and Birdie do not scale to large datasets (OOM)
- For indexing: STAR is $12\times$ to $393\times$ faster than TD systems
- For querying: Slower than Birdie on $\mathcal{D}_\ell$, but $16\times$ faster than the only scaling TD system (Pneuma) on $\mathcal{D}_f$

Experiments: Results Overview – Relevance@ k



Small dataset, reformulated questions



Large dataset, reformulated questions

Outline

- 1 Motivation: Open Statistical Data
- 2 Overview of Our Approach
- 3 Efficient Focused Web Crawling for Scalable Data Acquisition
- 4 Efficient and Scalable Search for Statistics
- 5 Conclusions and Perspectives

Conclusions

This thesis makes **open statistical data usable**, along a chain of problems:

- ① **Acquisition** → SB-CLASSIFIER, an RL crawler: **90% of targets in 20% of pages** [EDBT 2026] [VLDBJ under review]
- ② **Understanding** → a cell-type classification + table extraction pipeline, **scalable** and competitive with heavier models [CIKM 2026]
- ③ **Retrieval** → STAR, space- and time-aware discovery: **scales without a GPU** and **outperforms** heavier models [ICDE 2026] [TLDKS]
- ④ **Annotation** → TABAGREE, an **agreement-aware** platform for spreadsheet annotation [Demo to be submitted]

Two **public benchmarks** released (STATSHEETS, and the STD benchmark).

Efficient, scalable, and interpretable methods, on modest hardware.

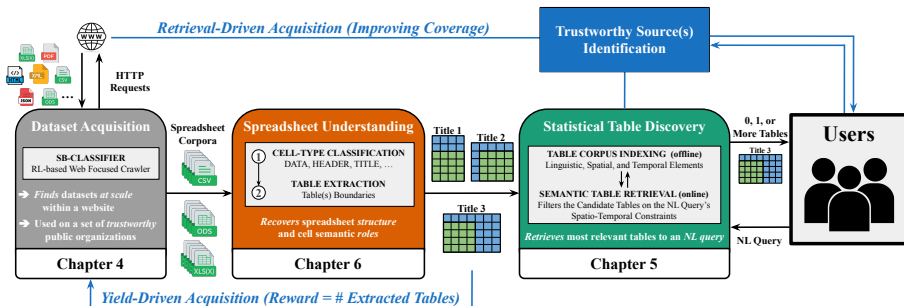
Each component opens its own directions:

- Acquisition
 - Reach the **hidden Web**
 - **Incremental re-crawling**, exploiting the learned structure
- Retrieval
 - **User feedback** into ranking
 - Efficient **statistical question answering** with small, local LLMs
- Understanding
 - Tighter **coupling with retrieval**, as its front-end
- Annotation
 - Make agreement **active** in the annotation process

Connecting the Tasks

The most compelling direction: **let the components feed one another.**

- **Yield-driven acquisition:** reward the crawler by how many usable tables spreadsheet understanding recovers downstream
- **Retrieval-driven acquisition:** when STAR detects **insufficient coverage** of a query, it triggers the acquisition of new tables



Thank you for your attention!

Any questions?

Bibliography I



Muhammad Imam Luthfi Balaka, David Alexander, Qiming Wang, Yue Gong, Adila Krisnadhi, and Raul Castro Fernandez.

Pneuma: Leveraging LLMs for Tabular Data Representation and Retrieval in an End-to-End System.

Proc. ACM Manag. Data, 3(3):200:1–200:28, 2025.



Lihu Chen, Gaël Varoquaux, and Fabian M. Suchanek.

Learning High-Quality and General-Purpose Phrase Representations.

In *Findings of the Association for Computational Linguistics, EACL 2024*, pages 983–994, 2024.



Muhammad Faheem and Pierre Senellart.

Adaptive Web Crawling Through Structure-Based Link Classification.

In *Proceedings of the 17th International Conference on Asia-Pacific Digital Libraries, ICADL 2015*, pages 39–51, 2015.

Bibliography II



Antoine Gauquier, Simon Ebel, Helena Galhardas, Théo Galizzi, Ioana Manolescu, Aurélien Peden, and Pierre Senellart.

Efficient and Scalable Search for Statistics.

In 41st Annual French Conference on Databases, BDA 2025, 2025.



Antoine Gauquier, Simon Ebel, Helena Galhardas, Théo Galizzi, Ioana Manolescu, Aurélien Peden, and Pierre Senellart.

Efficient and Scalable Search for Statistics.

In Proceedings of the 42nd IEEE International Conference on Data Engineering, ICDE 2026, pages 1113–1126, 2026.



Antoine Gauquier, Simon Ebel, Helena Galhardas, Théo Galizzi, Ioana Manolescu, Aurélien Peden, and Pierre Senellart.

STAR: Efficient, Scalable, and Modular Retrieval of Statistical Tables.

Transactions on Large-Scale Data- and Knowledge-Centered Systems (TLDKS), 2026.

Bibliography III



Yuxiang Guo, Zhonghao Hu, Yuren Mao, Baihua Zheng, Yunjun Gao, and Mingwei Zhou.

BIRDIE: Natural Language-Driven Table Discovery Using Differentiable Search Index.

Proc. VLDB Endow., 18(7):2070–2083, 2025.



Antoine Gauquier, Ioana Manolescu, and Pierre Senellart.

Efficient and Focused Web Crawling for Statistical Data Sources Retrieval.

In *2nd Workshop “Journées Infox sur Seine”*, 2024.



Antoine Gauquier, Ioana Manolescu, and Pierre Senellart.

Efficient Crawler for Scalable Web Data Acquisition.

In *40th Annual French Conference on Databases, BDA 2024*, 2024.

Bibliography IV



Antoine Gauquier, Ioana Manolescu, and Pierre Senellart.

Efficient Crawling for Scalable Web Data Acquisition.

In Proceedings of the 29th International Conference on Extending Database Technology, EDBT 2026, pages 372–385, 2026.



Antoine Gauquier, Ioana Manolescu, and Pierre Senellart.

Reinforcement Learning–Based Focused Web Crawling for Robust and Scalable Data Acquisition, 2026.

Under Review in VLDBJ.



Antoine Gauquier, Ioana Manolescu, and Pierre Senellart.

Structured Prediction for Scalable Spreadsheet Table Understanding: From Cell Types to Table Ranges.

In Proceedings of the 35th ACM International Conference on Information and Knowledge Management, CIKM 2026, 2026.



Antoine Gauquier, Ioana Manolescu, and Pierre Senellart.

Structured Prediction for Scalable Spreadsheet Table Understanding: From Cell Types to Table Ranges.

In 42nd Annual French Conference on Databases, BDA 2026, 2026.



Antoine Gauquier, Ioana Manolescu, and Pierre Senellart.

TabAgree: An Agreement-Aware Platform for Reliable Spreadsheet Table Annotation, 2026.

To be Submitted.



Antoine Gauquier, Ioana Manolescu, and Pierre Senellart.

TabAgree: An Agreement-Aware Platform for Reliable Spreadsheet Table Annotation.

In 42nd Annual French Conference on Databases, BDA 2026, 2026.



Andreas Kontogiannis, Dimitrios Kelesis, Vasilis Pollatos, Georgios Paliouras, and George Giannakopoulos.

Tree-based Focused Web Crawling with Reinforcement Learning, 2021.



Robert Kleinberg, Alexandru Niculescu-Mizil, and Yogeshwer Sharma.

Regret bounds for sleeping experts and bandits.

Mach. Learn., 80:245–272, 2010.



Robert Meusel, Peter Mika, and Roi Blanco.

Focused Crawling for Structured Data.

In *Proceedings of the 23rd ACM International Conference on Conference on Information and Knowledge Management, CIKM 2014*, pages 1039–1048, 2014.

Bibliography VII



Taylor Orth.

From millionaires to Muslims, small subgroups of the population seem much larger to many Americans.

<https://today.yougov.com/politics/articles/41556>, 2022.



Nils Reimers and Iryna Gurevych.

Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.

In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP 2019*, pages 3980–3990, 2019.

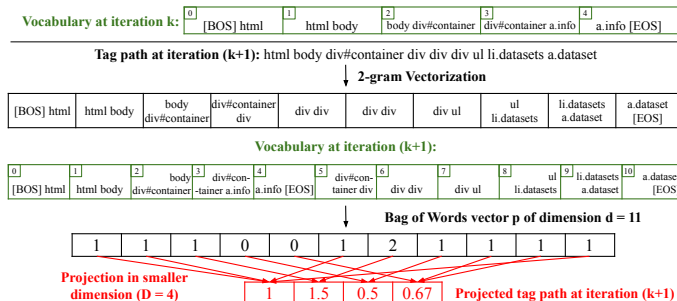


Qiming Wang and Raul Castro Fernandez.

Solo: Data Discovery Using Natural Language Questions Via A Self-Supervised Approach.

Proc. ACM Manag. Data, 1(4):262:1–262:27, 2023.

Building Groups of Hyperlinks – Merging strategy



Merging strategy: all groups are represented with a **centroid**. For each tag path:

- 1 Compute its set of (approx.) k -NN, and an associated **similarity score**
- 2 If the top-score $> \theta$ (threshold): **merge** and update the **centroid**
- 3 Otherwise: create a **new group**

Dataset Heterogeneity

Website size & target density:

- Website size ranges from 4k pages (qa) to $\gg$ 1M pages (ce, il, jp)
- Targets density ranges from 2.49% (in) to 66.78% (cl)

Target exposure:

- Low exposure: 0.7% (ok) $\Rightarrow$ targets linked from very few pages
- High exposure: 41% (is) $\Rightarrow$ nearly 1 out of 2 pages links to a target

Target depth:

- Shallow targets: $\sim$ 3 clicks (cl, qa)
- Deep targets: $>$ 60 clicks (in, ju)

Highly heterogeneous dataset in both content (topics, languages) and structure (size, density, exposure, depth)

Dataset: Selected Websites and Characteristics

Starting URL		Multilingual Fully Crawled	
<i>ab</i>	https://www.abs.gov.au/	X	X
<i>as</i>	https://www.assemblee-nationale.fr/	X	X
<i>be</i>	https://www.bea.gov/	X	✓
<i>ce</i>	https://www.census.gov/	X	✓
<i>cl</i>	https://www.collectivites-locales.gouv.fr/	X	✓
<i>cn</i>	https://www.cnis.fr/	X	✓
<i>ed</i>	https://www.education.gouv.fr/	X	✓
<i>il</i>	https://www.ilo.org/	✓	X
<i>in</i>	https://www.interieur.gouv.fr/	X	✓
<i>is</i>	https://www.insee.fr/	✓	✓
<i>jp</i>	https://www.soumu.go.jp/	✓	X
<i>ju</i>	https://www.justice.gouv.fr/	X	✓
<i>nc</i>	https://nces.ed.gov/	X	✓
<i>oe</i>	https://www.oecd.org/	✓	✓
<i>ok</i>	https://okfn.org/	✓	✓
<i>qa</i>	https://www.psa.gov.qa/ar/Pages/default.aspx	✓	✓
<i>wh</i>	https://www.who.int/	✓	X
<i>wo</i>	https://www.worldbank.org/	✓	X

Dataset: Selected Websites and Characteristics

	#Available (k)	#Target (k)	HTML with Tar. (%)	Target Depth
<i>ab</i>	952.26	263.26	8.86	8.94 ($\pm$ 2.56)
<i>as</i>	949.42	155.94	4.34	5.84 ($\pm$ 1.07)
<i>be</i>	31.23	15.84	32.19	5.73 ($\pm$ 3.21)
<i>ce</i>	988.37	257.68	3.47	4.23 ($\pm$ 0.48)
<i>cl</i>	5.54	3.70	5.40	2.80 ($\pm$ 0.82)
<i>cn</i>	12.80	7.49	13.87	4.26 ($\pm$ 1.59)
<i>ed</i>	102.71	10.47	3.95	11.89 ($\pm$ 13.22)
<i>il</i>	990.71	81.01	2.53	4.26 ($\pm$ 1.28)
<i>in</i>	922.46	22.98	1.54	66.94 ($\pm$ 39.43)
<i>is</i>	285.55	168.88	41.34	5.20 ($\pm$ 1.81)
<i>jp</i>	993.87	328.83	6.30	5.18 ($\pm$ 1.29)
<i>ju</i>	56.61	14.85	4.85	86.91 ($\pm$ 86.30)
<i>nc</i>	309.97	84.94	18.87	3.63 ($\pm$ 1.66)
<i>oe</i>	222.58	45.04	15.61	6.28 ($\pm$ 5.65)
<i>ok</i>	423.12	12.95	0.74	2.64 ($\pm$ 2.89)
<i>qa</i>	4.36	2.45	4.15	3.03 ($\pm$ 0.61)
<i>wh</i>	351.86	55.59	14.19	4.43 ($\pm$ 0.62)
<i>wo</i>	223.67	23.10	2.38	4.52 ($\pm$ 0.69)

- ① SB-CLASSIFIER: Our approach with URL classifier
- ② SB-ORACLE: Our approach with an **oracle** (perfect, costless URL classifier)
- ③ RANDOM: Random crawler
- ④ BFS: Breadth-First Search
- ⑤ DFS: Depth-First Search
- ⑥ FOCUSED: A standard focused crawler (focus = targets)
 - **Ablation** of our approach: neither tag path information, nor RL
- ⑦ TP-OFF: Offline-Trained, Tag Path-Based Crawler (adaptation of [FS15] to target retrieval)
 - **Ablation** of our approach: learning **offline** instead of online
- ⑧ TRES [KKP⁺21]: Topical, RL-Based Focused Crawler
 - Requires **substantial adaptations** to operate on target retrieval
 - Uses a Bi-LSTM (DL) classifier to assess **topic relevance**

URL Classifier Design

Evaluation metrics

- **Intra-website:** % of requests needed to reach 90% of targets
- **Inter-website:** Misclassification Rate (MR)

Feature sets

- **URL_ONLY:** URL features only
- **URL_CONT:** URL + anchor text + tag path + surrounding text

Models LR: Logistic Regression SVM: Support Vector Machine NB: Naive Bayes PA: Passive-Aggressive

Variant	<i>be</i>	<i>cl</i>	<i>cn</i>	<i>ed</i>	<i>in</i>	<i>is</i>	<i>ju</i>	<i>nc</i>	<i>oe</i>	<i>ok</i>	<i>qa</i>	MR
URL_ONLY-LR	82.1	75.1	71.3	53.2	11.7	76.1	36.5	52.6	60.7	15.9	62.3	2.62
URL_ONLY-SVM	82.7	75.7	71.8	63.6	11.3	76.0	37.4	52.2	63.5	16.7	61.5	2.99
URL_ONLY-NB	82.9	75.2	72.1	53.7	11.4	76.3	35.8	52.7	59.7	18.0	63.1	2.92
URL_ONLY-PA	82.3	74.4	71.7	53.3	11.1	75.8	36.7	51.6	60.5	15.9	60.9	2.56
URL_CONT-LR	82.2	74.4	71.9	54.3	11.3	76.4	37.8	52.9	64.7	16.8	60.0	5.93
URL_CONT-SVM	82.6	75.0	71.8	52.8	11.6	76.4	38.8	53.1	61.1	18.7	60.1	6.36
URL_CONT-NB	84.1	74.7	71.9	53.6	11.4	75.7	35.5	52.3	59.9	19.1	60.4	7.15
URL_CONT-PA	82.5	75.1	71.9	53.6	11.6	76.2	38.4	52.1	62.6	16.1	60.6	4.12

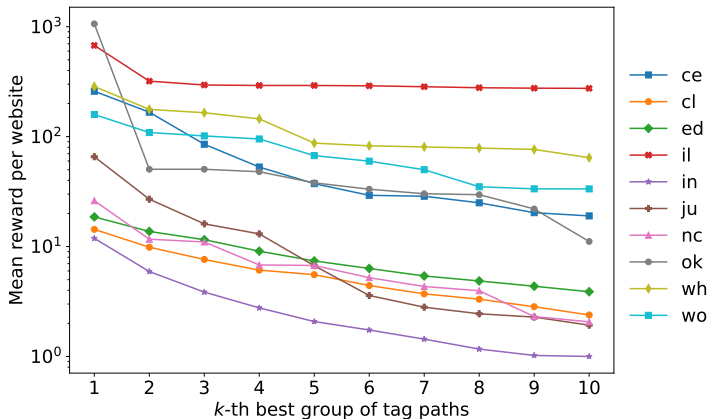
URL Classifier Results (Overview)

[Confusion matrix](#) of our URL classifier, on the fully-crawled websites (average over 15 runs)

True \ Predicted			
	HTML (%)	Target (%)	Neither (%)
HTML	58.04	1.37	0.00
Target	0.75	32.19	0.00
Neither	5.34	2.41	0.00

Efficiency of Sleeping Bandit Learning

Mean rewards of the **top-10 groups of tag paths** for ten websites



- The agent **efficiently learns** the fruitful groups
- **Empiric validation** of our research hypotheses

SD Retrieval Across Sample Targets

Experimental setup

- Random sample of $7 \times 40 = 280$ targets
- 7 diverse websites

	<i>be</i>	<i>ed</i>	<i>is</i>	<i>in</i>	<i>nc</i>	<i>oe</i>	<i>wh</i>
SD Yield (%)	82	35	93	40	83	60	40
Mean # SDs / Target	9.1	2.8	2.9	2.1	2.1	4.9	1.4

- SDs are present in a substantial fraction of targets across all websites
- Even non-statistical websites show notable SD presence
- SDs are concentrated: when present, typically > 2 SDs per target

Relaxation Mechanism

What if **no table matches** s_q and t_q exactly?

→ Look up **neighbors** at increasing distance d in H_S/H_T

E.g.: In 2025, what was the fertility rate in *Yvelines*? With $N = 3$ results

$d = 0 \rightarrow$ constraints = (Yvelines, 2025). No matching tables in H_S and H_T :

0 match $< N$: keep **relaxing**

↓ $pen = d \cdot p$

$d = 1$, relax **space**: Yvelines $\rightarrow \{\hat{\text{Île-de-France}}\}$. **New constraint**:

($\hat{\text{Île-de-France}}$, 2025)

New ANN lookup. Scores: $\{(T_1, 0.65)\}$

1 match $< N$: keep **relaxing**

↓ $pen = d \cdot p + d \cdot p = 2 \cdot d \cdot p$

$d = 1$, relax **space and time**: 2025 $\rightarrow \{2024, 2026\}$. **New constraints**:

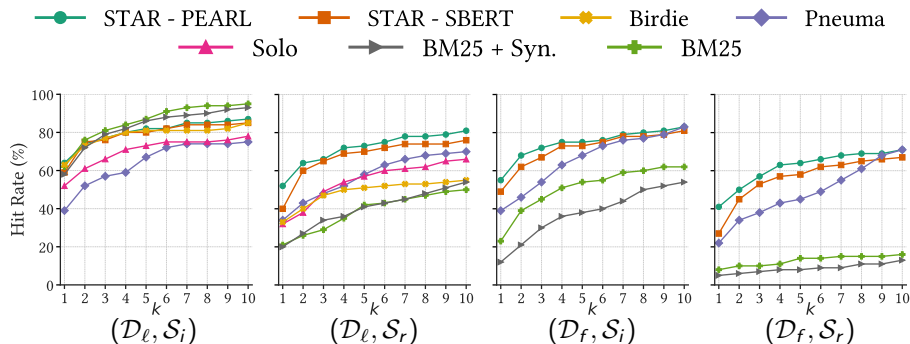
$\{\hat{\text{Île-de-France}}\} \times \{2024, 2026\} \Rightarrow (\hat{\text{Île-de-France}}, 2024), (\hat{\text{Île-de-France}}, 2026)$

New ANN lookup. Scores: $\{(T_2, 0.85), (T_3, 0.50)\}$

3 matches $\geq N$: stop. **Final ranking**: T_2, T_1, T_3

$$\text{Score}(T) = \sum_{x_T \in T} \text{sim}(\phi(\bar{q}), \phi(\overline{x_T})) - pen$$

HitRate@k Results



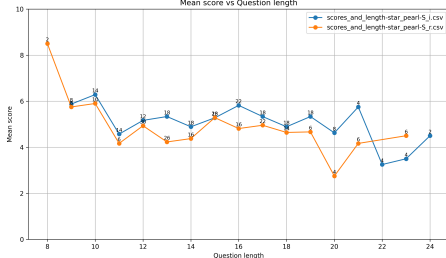
On $(\mathcal{D}_\ell, \mathcal{S}_i)$: **BM25 (+Syn.) leads**, near-perfect at $k=10$

On the **three other settings**, STAR is (at least) as good as PNEUMA

Does Question Length Hurt STAR-PEARL?

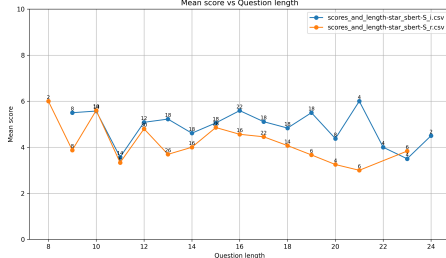
STAR-PEARL

Mean score vs Question length



STAR-SBERT

Mean score vs Question length

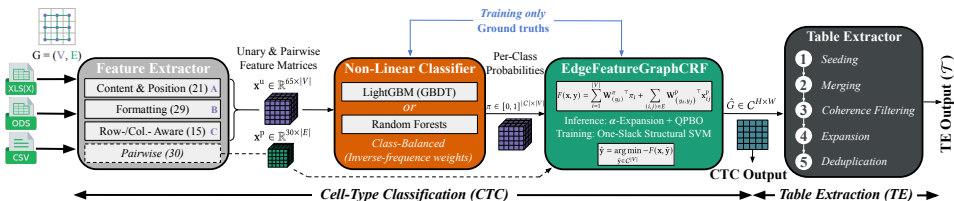


Mean Relevance@5 vs. question length (in words)

No degradation trend for PEARL

→ Despite being trained on short, context-free phrases

Spreadsheet Understanding Pipeline



- **Input:** Raw grid of cells (content, formatting, position) from sheets
- **Outputs:**
 - 1 Table ranges (e.g., A5:D29)
 - 2 Type of each cell (DATA, HEADER, EMPTY, TITLE, OTHER)

No learning in TE, **no GPU** anywhere in the pipeline

Table Extraction: Hyperparameters

Interpretable pipeline: 12 hyperparameters, **set once**

→ No cross-validation, unlike CTC classifiers (LIGHTGBM, RF, CRF)

Param.	Val.	What it is	Why
$n_D^{\min}$	4	Min. DATA cells to seed	Avoids tiny seeds from stray numbers
$n_H^{\min}$	8	Min. HEADER cells to seed	Header blocks need enough width
τ_H	0.4	Header-density threshold (rows)	Separates header rows from noise
τ_H^c	0.4	Header-density threshold (cols)	Same for column-oriented headers
g_v	4	Min. sep. rows between 2 seeds	Small gaps shouldn't split a table
g_h	2	Min. empty cols between 2 seeds	Same logic, horizontally
$\rho_D^{\min}$	0.05	Min. DATA density in box	Rejects mostly empty padding
$\rho_H^{\min}$	0.5	Min. HEADER density in box	Header bands are mostly header cells
$n_D^{\dagger}$	20	Threshold for header-less seeds	Lets dense DATA blocks survive
r_N	3	Header search radius	Recovers headers slightly detached
τ_{IoU}	0.5	IoU threshold (dedup.)	Merges near-duplicate detections
τ_{cont}	0.8	Containment threshold (dedup.)	Drops a box inside another

No tuning, yet competitive with TUTA and SpreadsheetLLM

→ Human-like heuristics are strong on human-challenging tasks

The STATSheETS Dataset

Scale:

- **737 sheets** from 14 public organizations, 818 annotated tables
- **7 570 354 cells** annotated (avg. 10 272 per sheet)

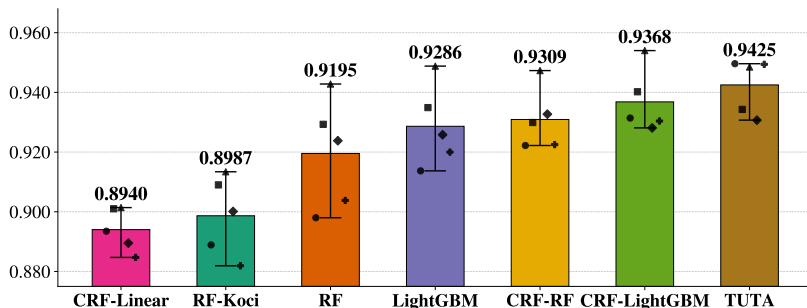
Diversity:

- **6 languages**: English, French, Arabic, Japanese, Spanish, Icelandic
- **5 formats**: XLSX (42.9%), XLS (40.2%), ODS (8.1%), CSV (7.1%), TSV (1.8%)

Class imbalance (per cell): **DATA** 62.2% – **HEADER** 20.4% – **EMPTY** 17.1% – **TITLE/OTHER** < 1% each

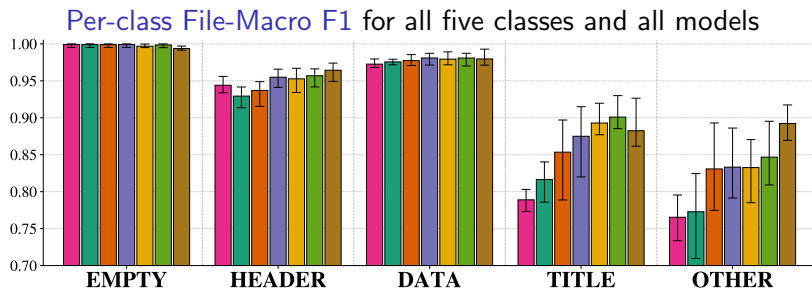
CTC: Ablation and Comparison to TUTA

Mean File-Macro F1 scores per model for all five folds



- Spatial features (RF-KOCI→RF): **+2.1 pts**
- Sequential + non-linear (RF→CRF-LIGHTGBM): **+1.7 pts**
- CRF-LIGHTGBM (0.937) is within **0.6 pts** of TUTA (0.943, **Transformer, GPU, semantic-dependant**)

CTC: Per-Class Breakdown



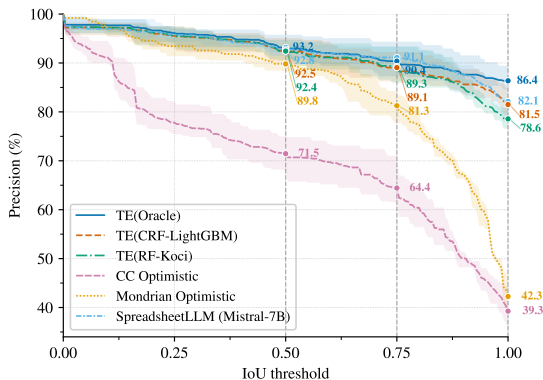
TUTA vs. our pipeline:

- Comparable on HEADER/DATA/EMPTY
- TUTA ahead on **OTHER** (+4.5 pts, **most semantic-dependant class**)
- Our pipeline leads on **TITLE** (+1.9 pts, more structural)

Table Extraction: Macro Precision

Macro precision: per-sheet precision curve, then **averaged across sheets**

→ Every sheet counts equally, regardless of how many tables it contains

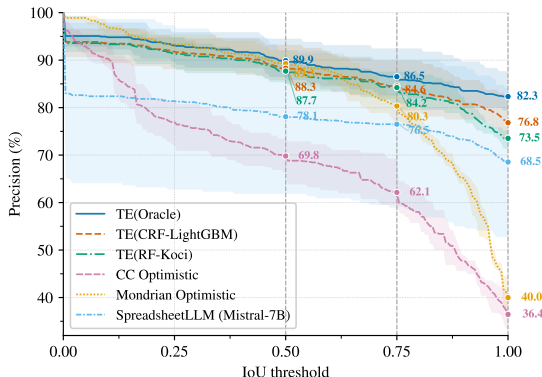


- TE(CRF-LIGHTGBM) is not far from TE(ORACLE) (+4.9 pts left)
- SPREADSHEETLLM is slightly ahead (+0.6 pt)
- MONDRIAN (generic region detection) is far behind: regions $\neq$ tables

Table Extraction: Micro Precision

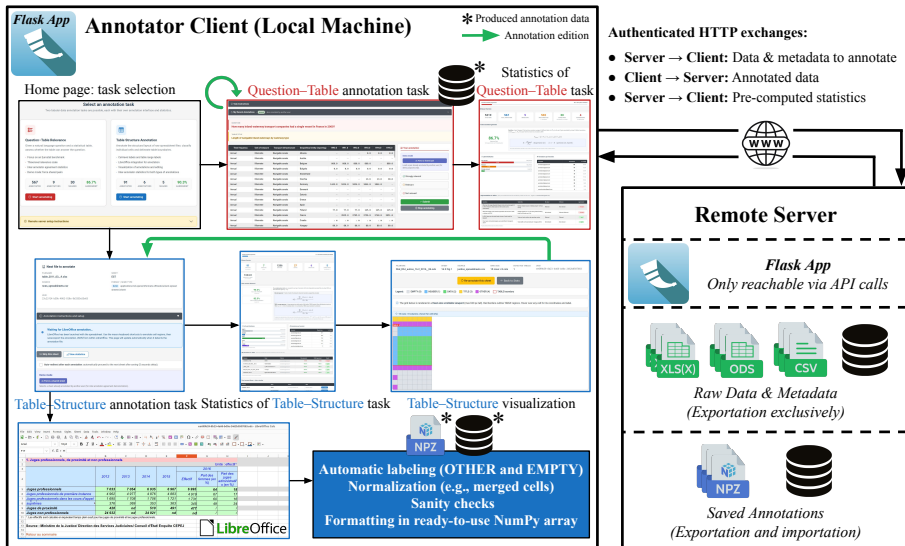
Micro precision: predictions **pooled across sheets** before curve computation

→ **Sheets with more tables weigh more**



- Macro > micro for **every** method
- SPREADSHEETLLM **drops sharply** (−8.3 pts vs. CRF-LightGBM)
 - **SpreadsheetLLM over-predicts** (bad at *only* returning the right ones)
 - and costs >100× **more** to run

TABAGREE: Architecture



TABAGREE: Agreement Metrics

Question–Table relevance (multi-class, one label per pair):

$$A = 1 - \frac{|P_{\text{disagree}}|}{|P_{\geq 2}|}$$

→ Proportion of multi-annotated pairs *without* disagreement

Table–Structure (per-cell labels *and* table ranges)

For annotators a, b on the same sheet:

$$A_{a,b}^{\text{cell}} = \frac{\sum_{(r,c)} \mathbb{1}[\ell_a(r, c) = \ell_b(r, c)]}{|R_a \cap R_b|} \quad A_{a,b}^{\text{tbl}} = \frac{\sum_{(r,c)} \mathbb{1}[t_a(r, c) = t_b(r, c)]}{|R_a \cap R_b|}$$

Directly interpretable as a proportion